303-Architecture and Town Planning

Islamic Architecture

Overview

- What is Islamic Architecture?
- Basic Principles of Islamic Architecture
- Types of Islamic Architecture
- Elements of Islamic Architecture
- Styles of Islamic Architecture
- Islamic decorative elements

Islamic Architecture

Definition

Islamic architecture can be defined as a building traditions of Muslim populations of the Middle East and any country where Islam has been dominant from the 7th century onwards.

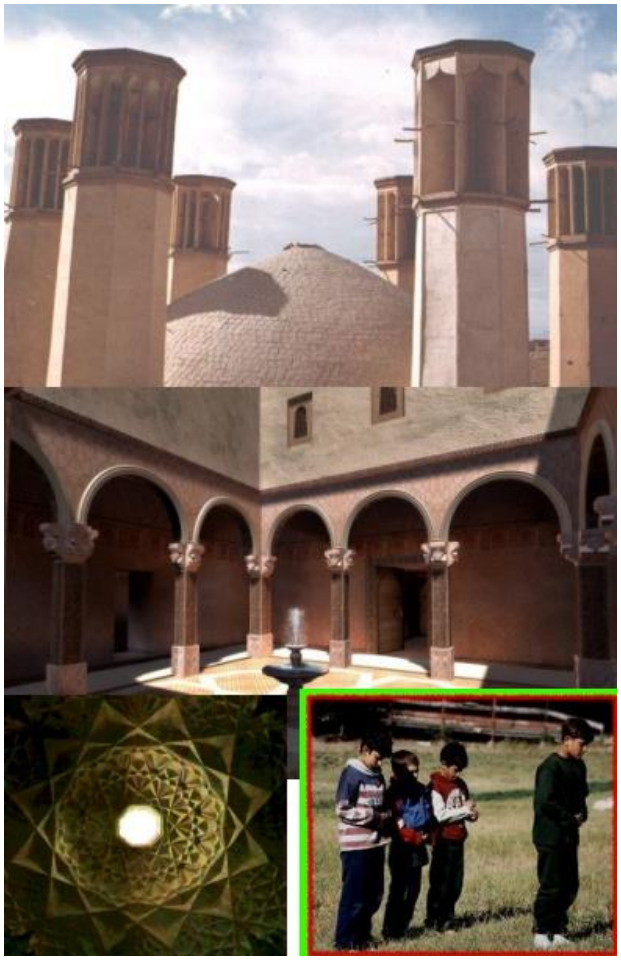
Encompasses both secular & religious buildings from the beginning of Islam to present day, influencing the design & construction of structures within the sphere of Islamic Culture

Common Interpretations of Islamic Architecture

- The concept of Allah's infinite power is evoked by designs with repeating themes.
- Human and animal forms are rarely depicted in decorative art as Allah's work is matchless.
- Calligraphy is used to enhance the interior of a building by providing quotations from the Qur'an.
- Islamic architecture focuses on the beauty of the interior rather than exterior spaces.
- Use of impressive forms such as large domes, towering minarets, and large courtyards are intended to convey power

Basic Principals of Islamic Architecture

- **Concept of Space**
  - Sacred
- **Concept of Man on Earth**
  - Life Cycle of a Grave
- **Geometric Drivers**
  - Patterns in Nature Sufi Elements on Design
- **Environmental Constraints**
  - Form addresses climate Forms follow Function or visa versa?



Types of Islamic Buildings

- **Mosque – Muslim center @ place for worship**
- Madrasah – public school
- Hammam - A structural design for a hot bathhouse
- Caravanserai - A roadside inn for travellers
- Casbah/Citadel - a fortress
- **Mausoleum - a tomb or a monument**

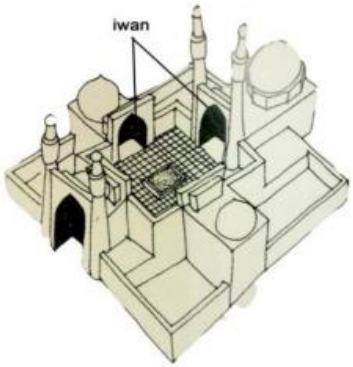








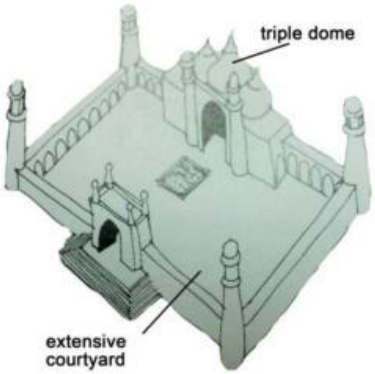
Mosque Typology



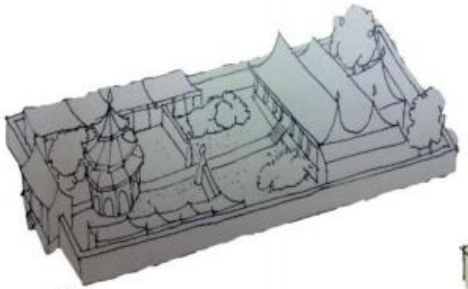
Iran and central Asia:
The bi-axial four iwan type



Anatolia:
Use of massive central dome

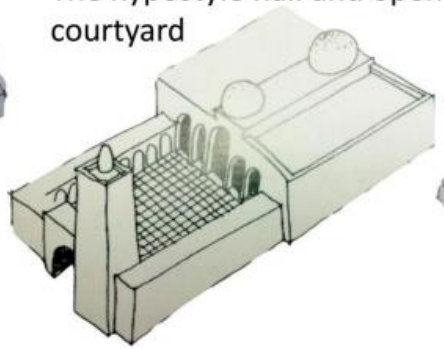


The Indian subcontinent:
Triple domes and an extensive courtyard

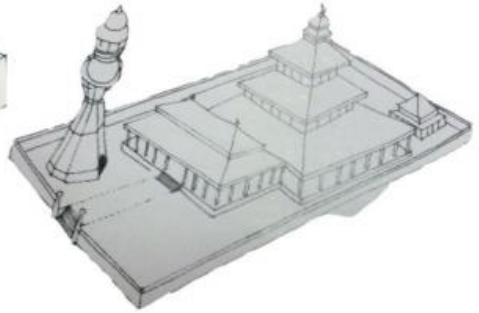


China:
Detached pavillions within a Walled garden enclosure

Spain, North Africa:
The hypostyle hall and open courtyard



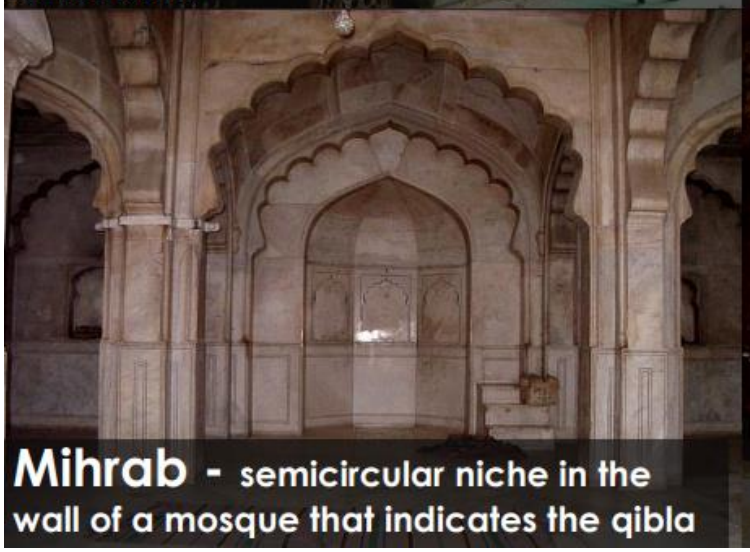
South-East Asia:
Central pyramidal roof construction



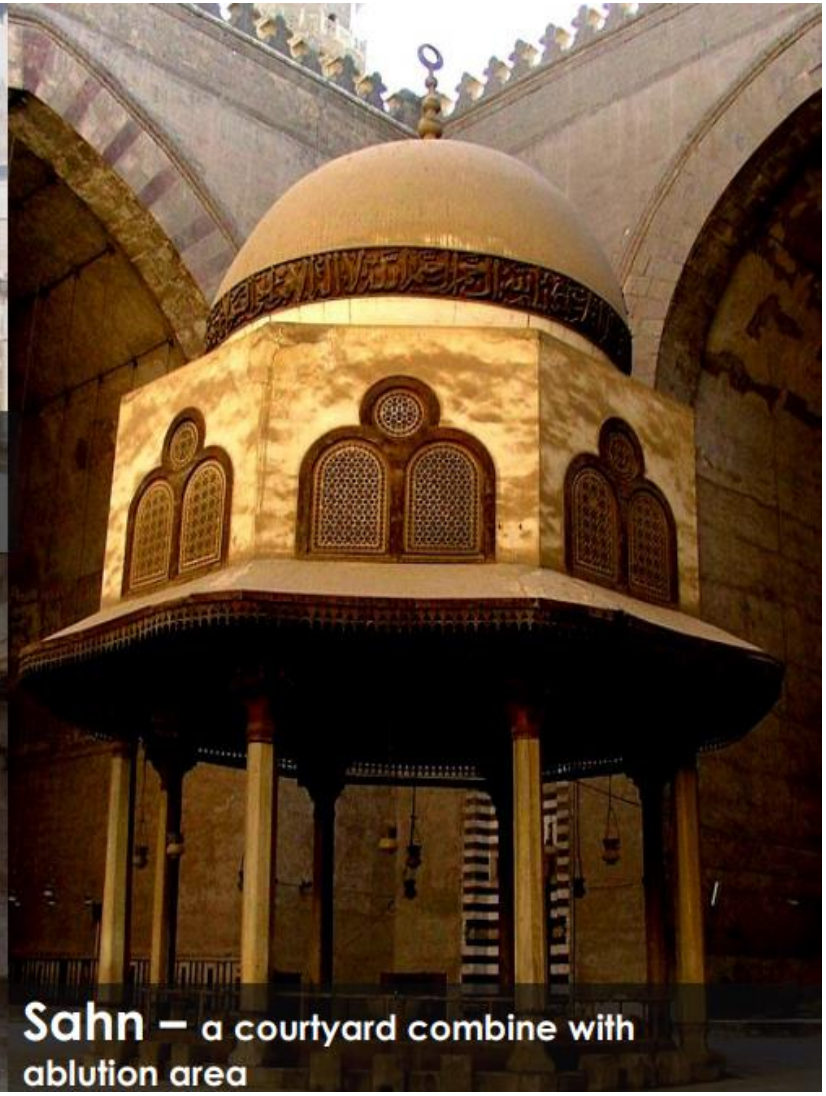
Mosque Architecture Features



Mashrabiya – projecting window enclosed with carved wood latticework



Mihrab - semicircular niche in the wall of a mosque that indicates the qibla

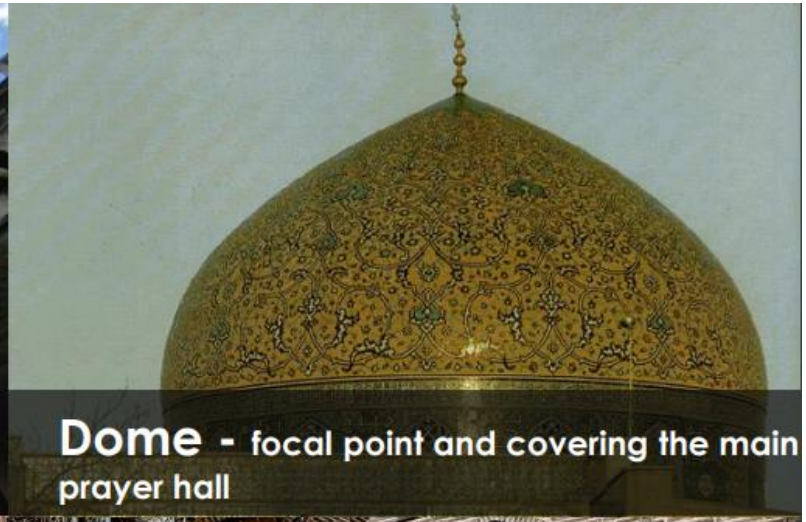


Sahn – a courtyard combine with ablution area

Mosque Architecture Features



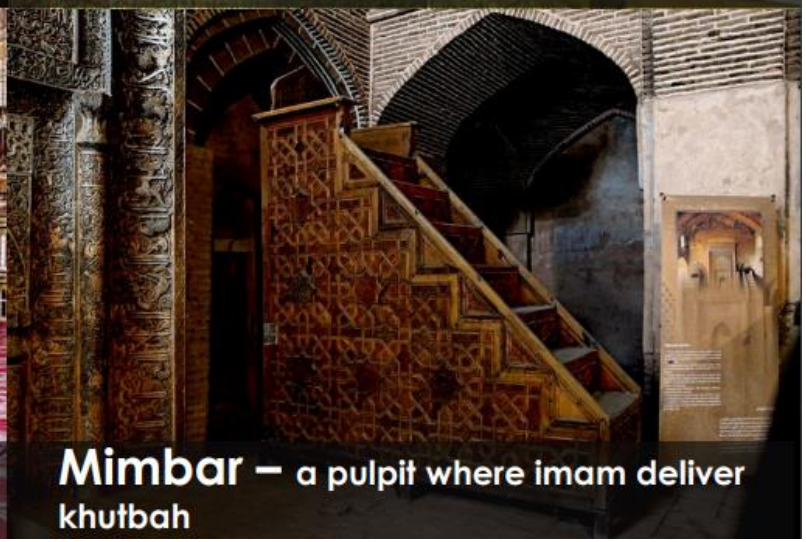
Minaret - visual focal point and are used for the call to prayer



Dome - focal point and covering the main prayer hall



Dikka – a place of delivering royal message



Mimbar – a pulpit where imam deliver khutbah

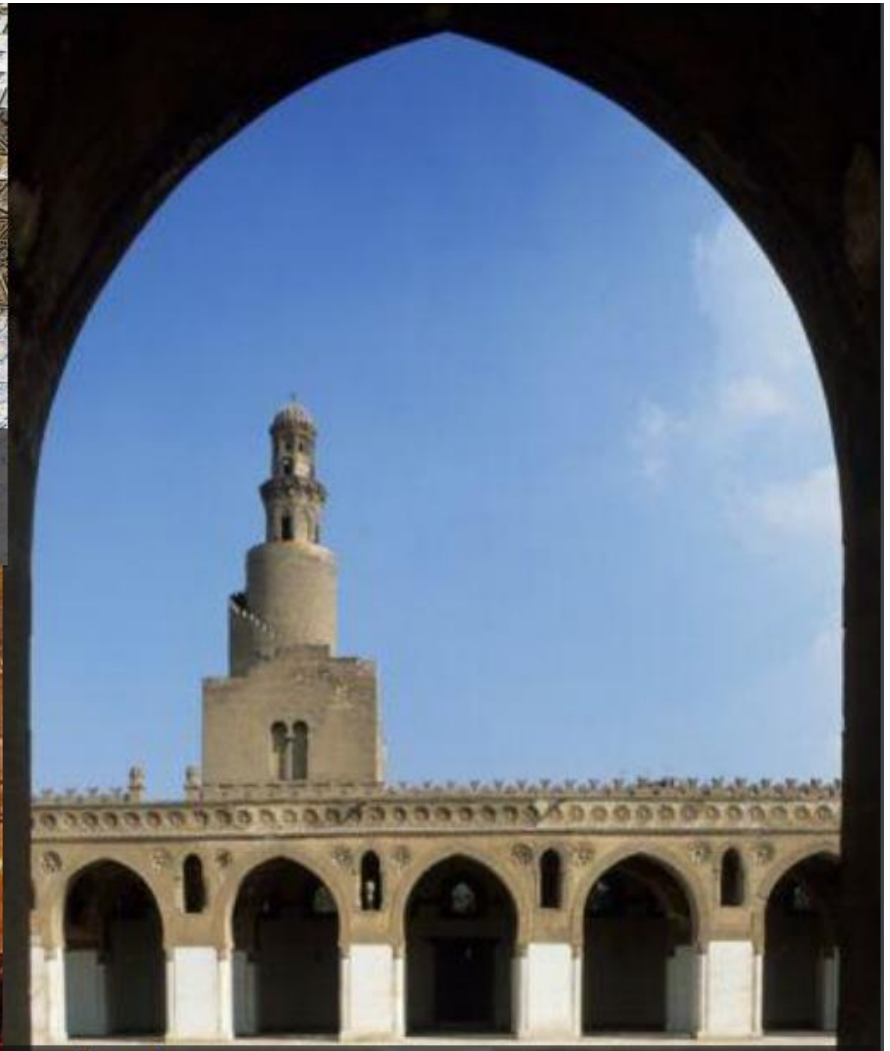
Mosque Architecture Features



Muqarnas - stalactite vault, Persian architectural ornament



Praying area hall – main area to perform congregational pray

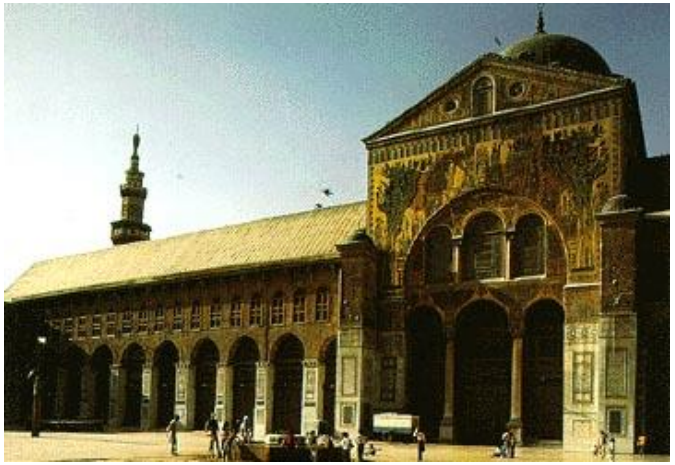


Arch – structure that spans a space while supporting weight

Great Mosque of Damascus

Damascus, Syria

- The interior walls of **Great Mosque of Damascus** have stone mosaics that depict crowns, fantastic plants, realistic trees, and even empty towns.
- This is thought to represent Paradise for the faithful Muslim.
- Both the Dome of the Rock in Jerusalem and **The Great Mosque of Damascus** used the Syrian cut-stone technique of building and popularized the use of the dome.

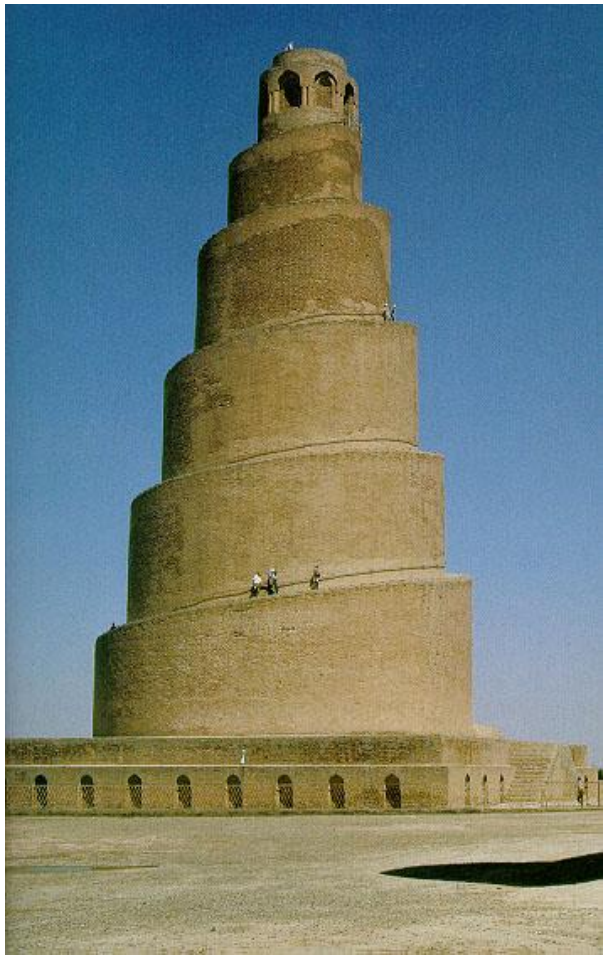




The Great Mosque of Samarra

Samarra, Iraq

In 750 the **Abbasid Dynasty** moved the capital east to Baghdad, and from 836 to 892 the Abbasid rulers resided at Samarra. **The Great Mosque of Samarra** is an important example of the Iraqi style, noted for its massive size and huge minaret.



Mosque of Sultan Hassan (Cairo, Egypt)



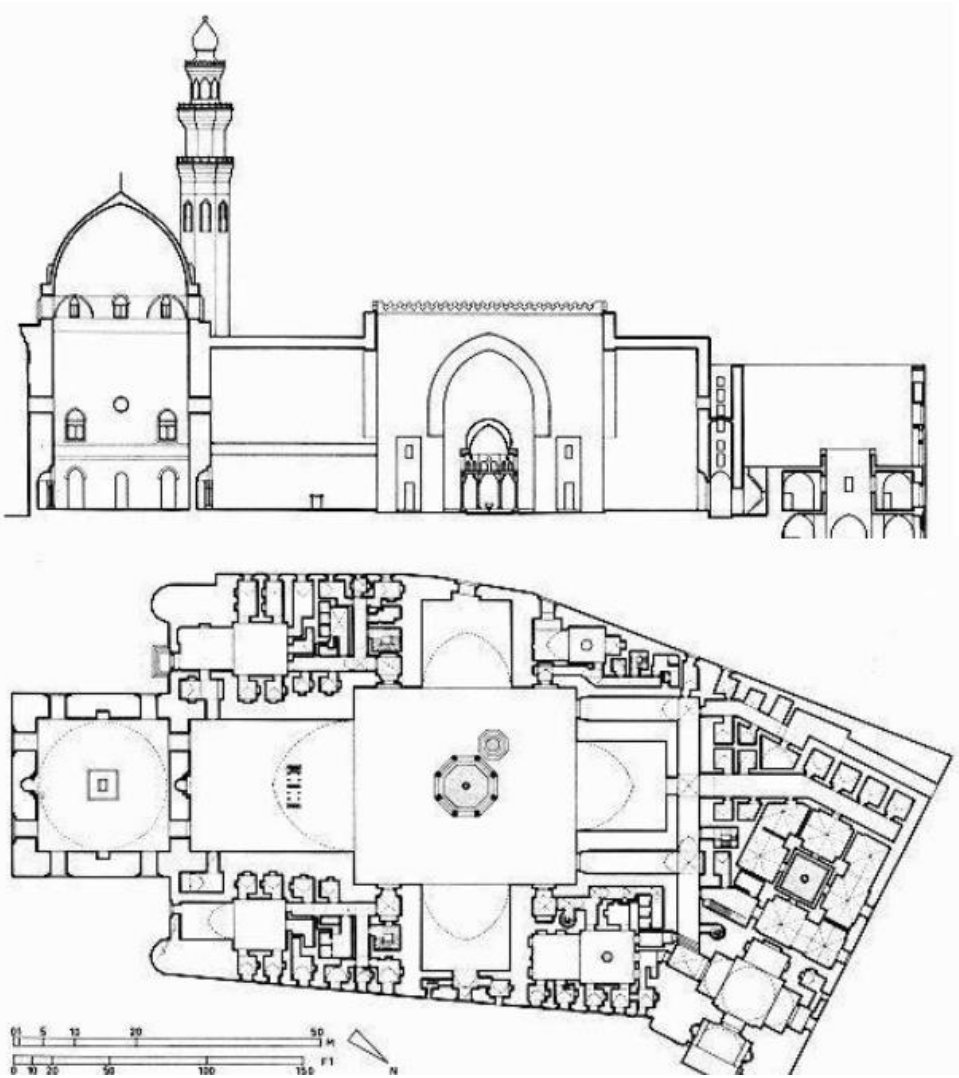
Background History

- The construction of the building started in 1356 AD by Sultan Hassan and finished in 1363 AD by Basyir Aga, one of his prince.
- Historians believe the mosque used stone from the pyramids at Giza.
- One of the minarets collapsed during construction killing 300 people.

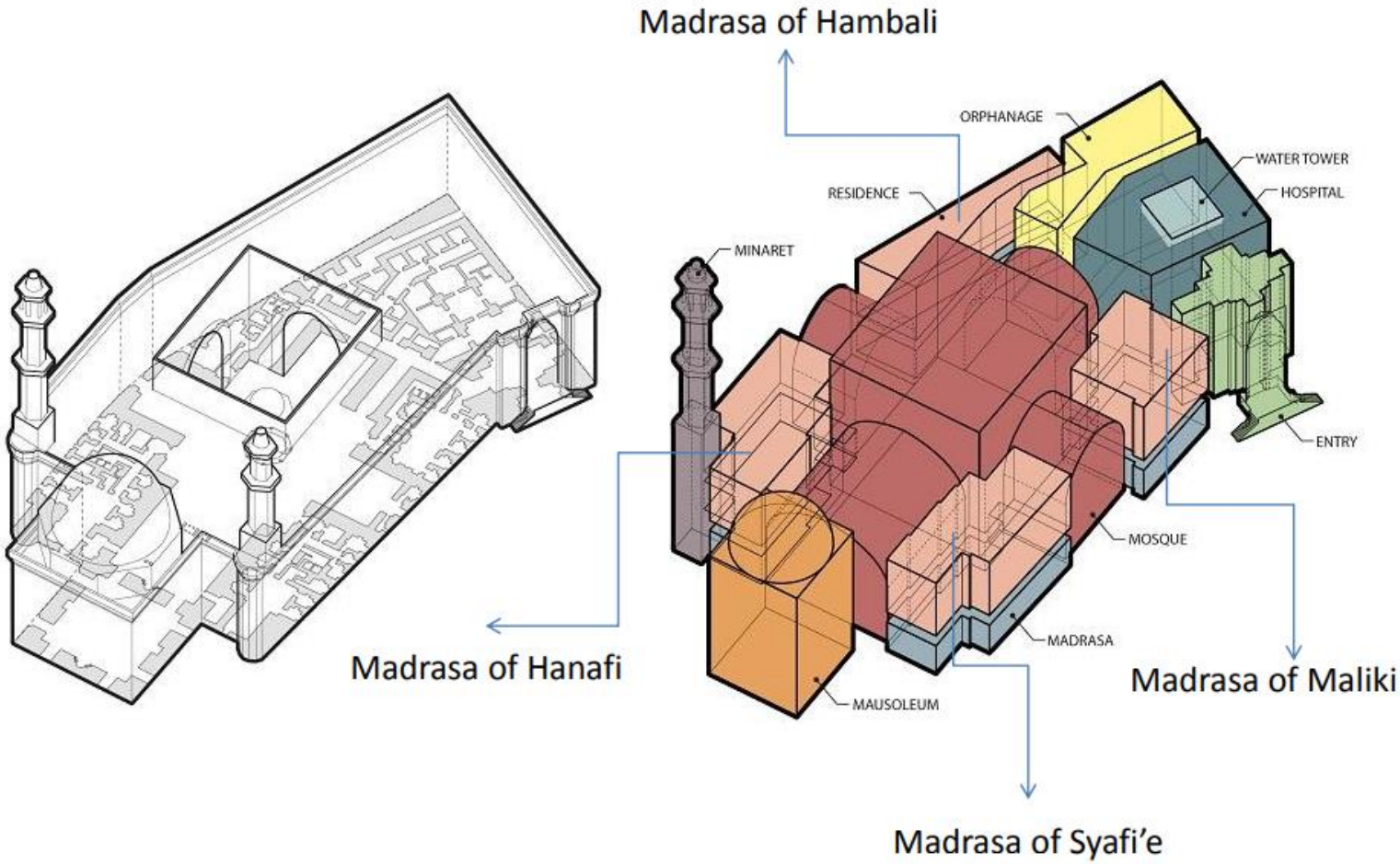
Mosque of Sultan Hassan (Cairo, Egypt)

Characteristics

- the building become a mosque and religious school for all four juristic branches of Sunni Islam – Shafi'e, Maliki, Hanafi and Hambali.
- represent great Mamluk architecture monument in Cairo.
- The facade is about 76 meters long and 36 meters high.
- The cornices, entrance portal with pointed arch, burial chamber, and the monumental staircase are particularly noteworthy.
- Verses from the Quran in elegant Kufic and Thuluth scripts adorn the inner walls.



Mosque of Sultan Hassan (Cairo, Egypt)



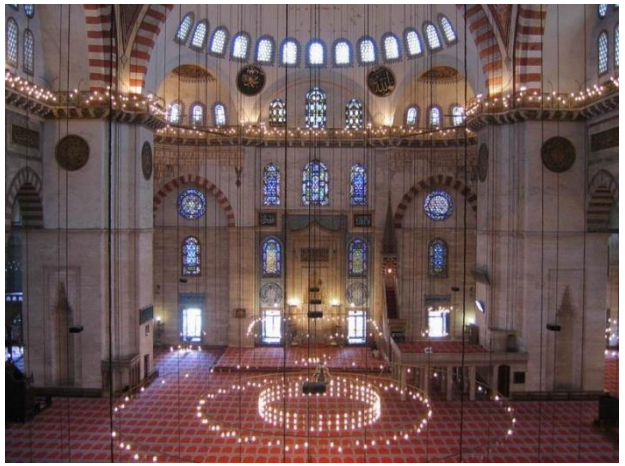
Mosque of Sultan Hassan (Cairo, Egypt)



Sulayman Mosque Complex

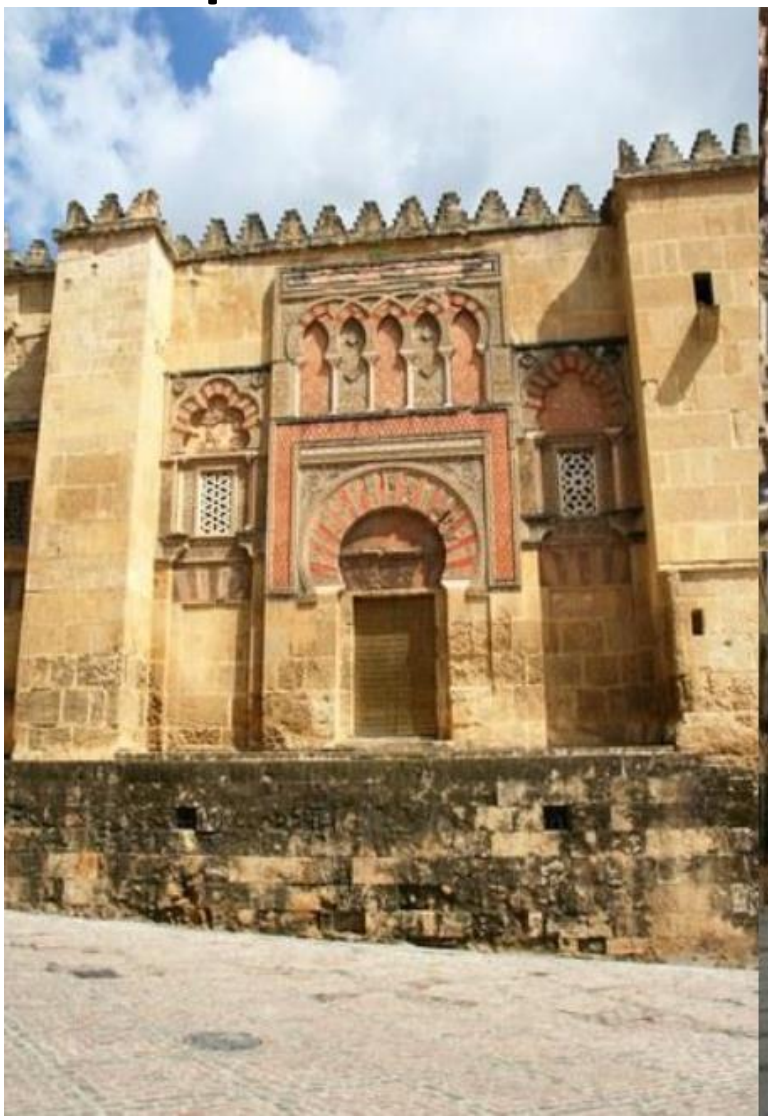
Istanbul, Turkey





- In Turkey the mosque form was also derived from Persia, as was most Turkish art.
- One of the most famous Turkish architects was Sinan, chief architect in the Ottoman court from 1539 until his death in 1588. He constructed or designed most of Sulaymani's buildings, the most noted of which is his **Sulayman Mosque** (c.1557) in Istanbul.

Mosque of Cordova (Andalusia, Spain)







Mosque of Cordova (Andalusia, Spain)

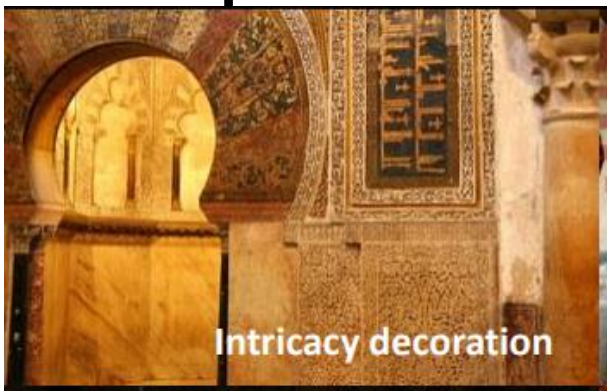
Characteristics:

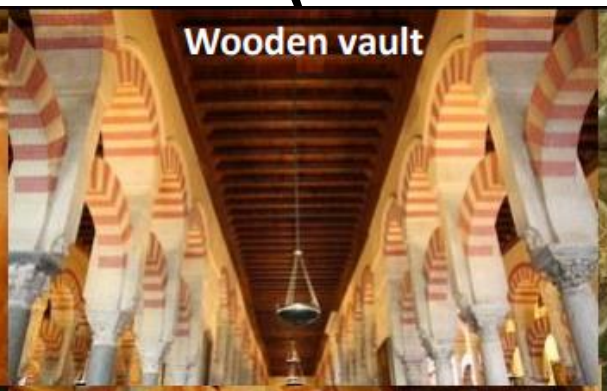
- **Giant arches** and its forest of over 856 (of an original 1,293) **columns** of jasper, onyx, marble, and granite. These were taken from the Roman temple which had previously occupied the site and other destroyed Roman buildings.
- The **Torre del Alminar**, the minaret once used to summon the faithful to prayer, has a Baroque tower.
- The Great Mosque of Cordova was begun in 785 and is famous for its rows of double-tiered arches.

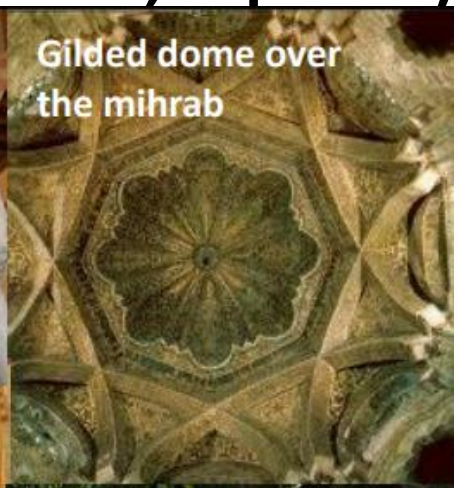
Mosque of Cordova (Andalusia, Spain)



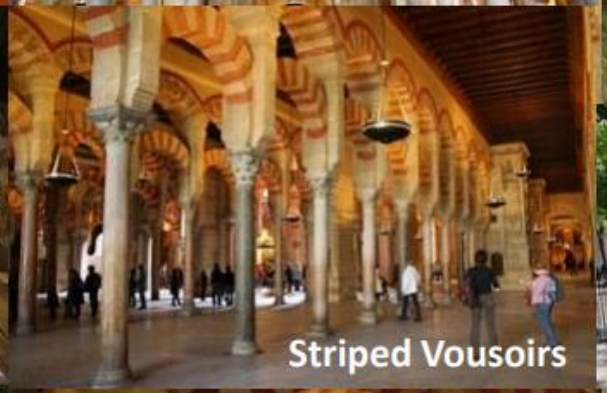
Mosque of Cordova (Andalusia, Spain)















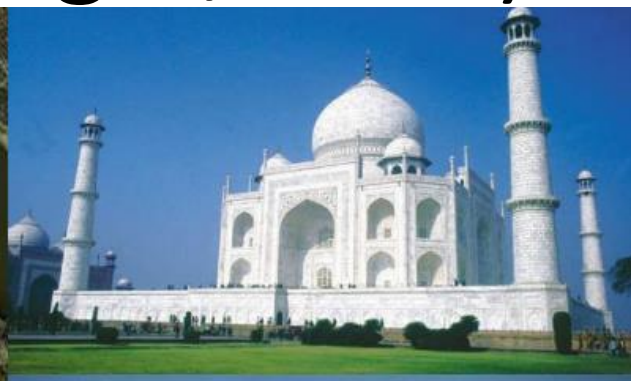


The Taj Mahal (Agra, India)

- **Taj Mahal** was built in 22 years (1631-1653) with the orders of Shah Jahan and it was dedicated to Mumtaz Mahal (Arjumand Bano Begum)
- 20,000 workers labored and 32 crore rupees were spent during the construction of the monument
- The Architect of Taj Mahal was Ustad Ahmad Lahauri
- **Taj Mahal** means the' Palace of the Crown' because; Taj means "Crown" and Mahal means "palace".
- The **Taj Mahal** incorporates and expands on many design traditions, particularly Persian, Hindu and earlier Mughal architecture.

The Taj Mahal (Agra, India)



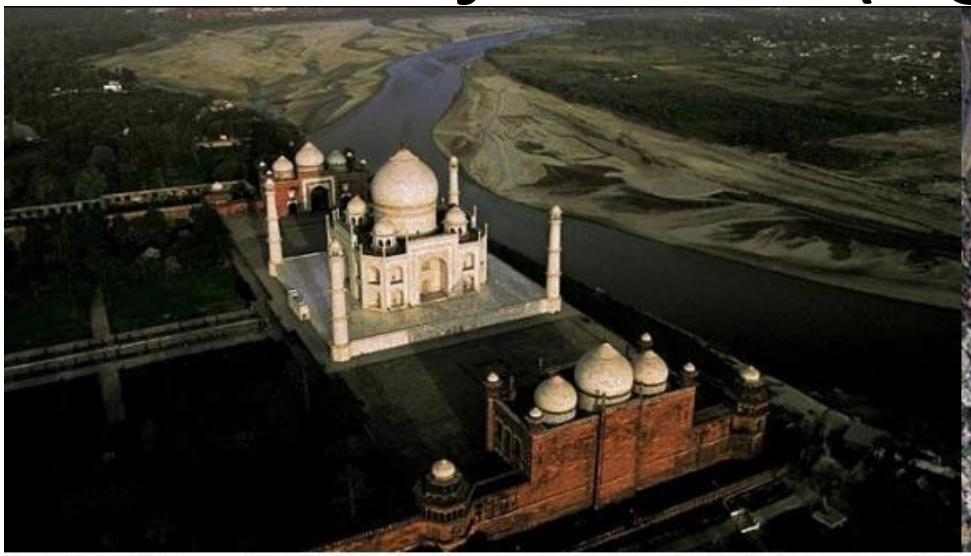




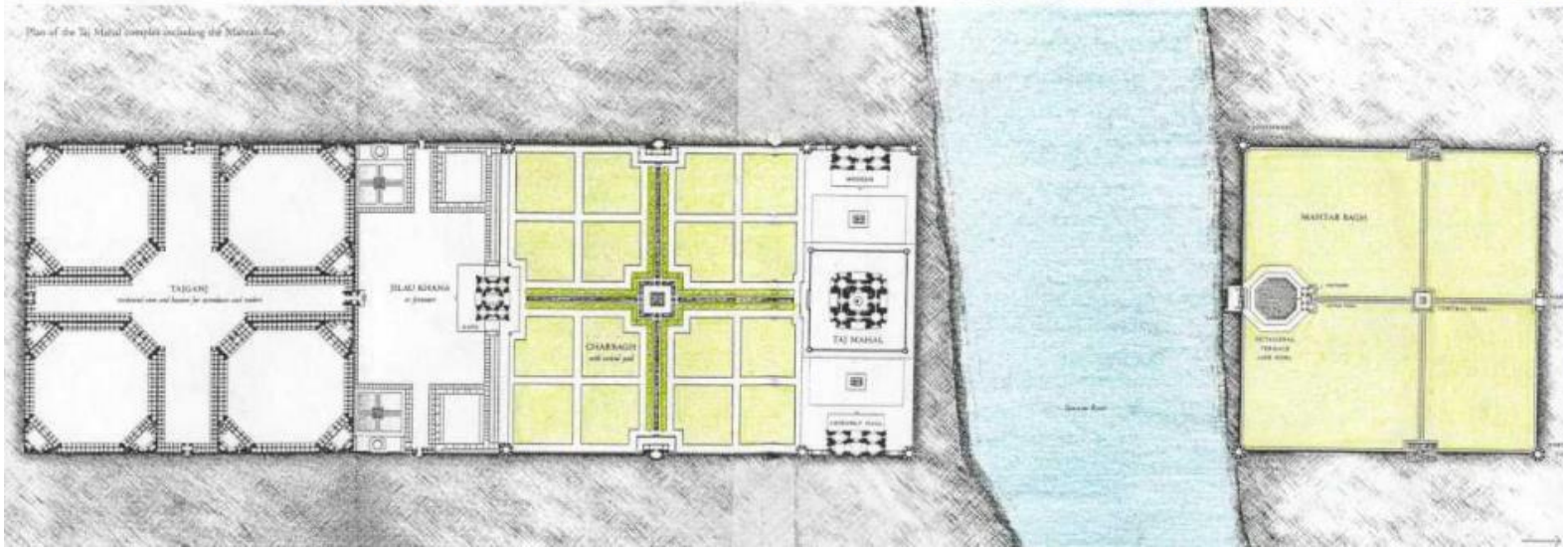




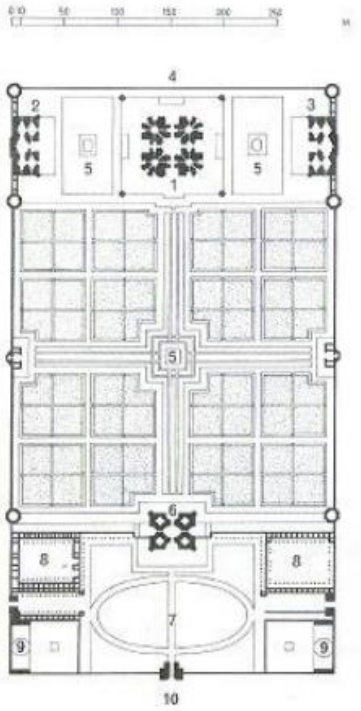
The Taj Mahal (Agra, India)

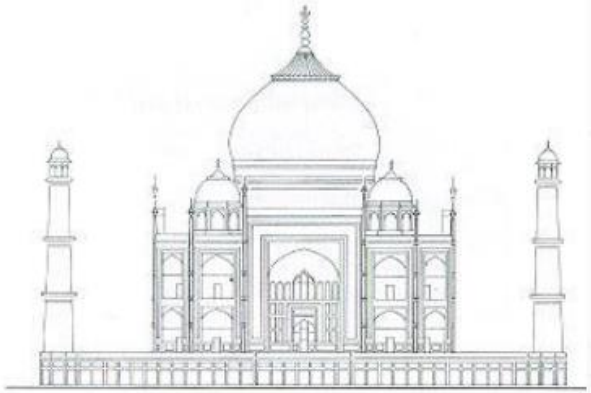


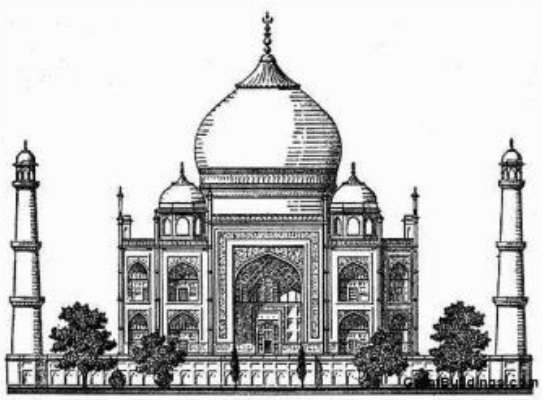


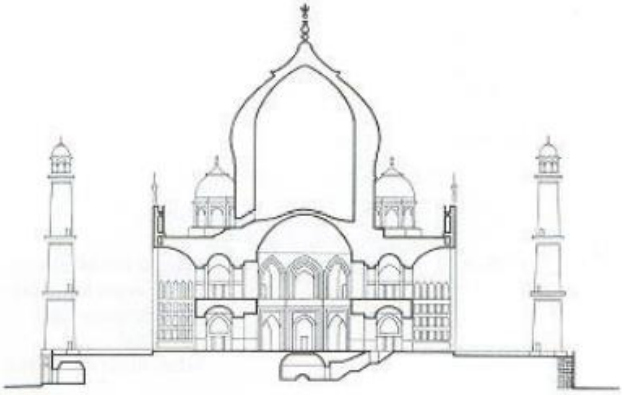


The Taj Mahal (Agra, India)









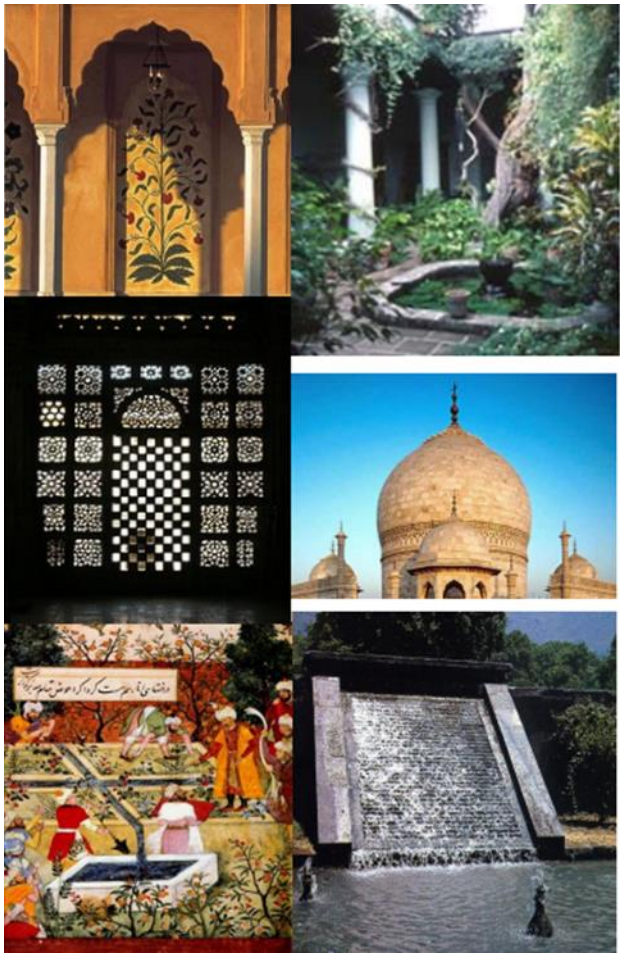


1. Tomb
2. Mosque
3. Guest house
4. Yamuna River
5. Ornamental pool
6. Main Gate
7. Forecourt
8. Servants' quarters
9. Royal tombs
10. Mumtazabad (Taj Ganj)

Elements of Islamic Architecture

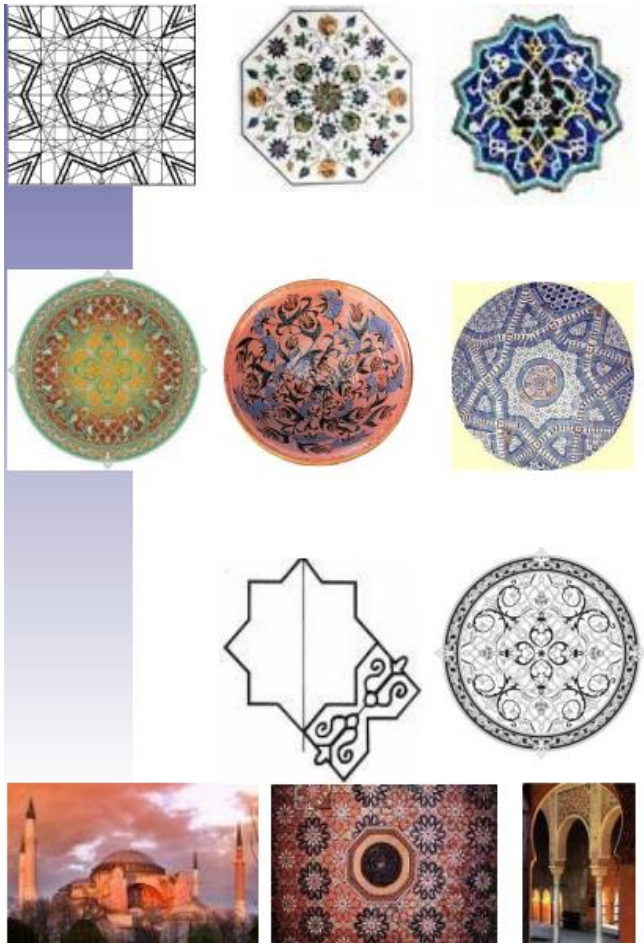
Islamic architecture are identified with the following design elements, which were inherited from the first mosque built by Muhammad in Medina, as well as features adapted from churches and other religious places.

- **Minarets** or towers
- **Mihrab** or niche on an inside wall indicating the direction of Mecca
- **Domes/Arches**
  - Use of geometric shapes and repetitive art (arabesque)
  - Use of decorative Arabic calligraphy
  - Ablution fountains
- **Courtyard** – Sehan
- **Garden** - Water Char Bagh
- **Bara Dari**
- **Awan**



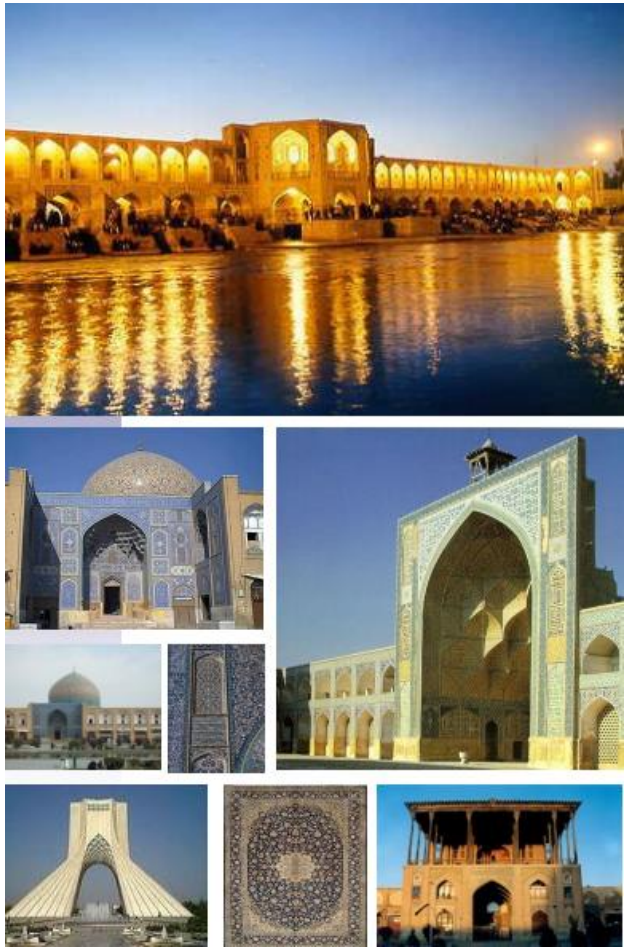
Styles of Islamic Architecture

- Persian Moorish
- Turkistan (Timurid)
- Ottoman Turkish
- Fatimid
- Mamluk
- Indo-Islamic (Mughal)
- Sino-Islamic
- Afro-Islamic



Persian - Isfahan

- Early Islamic architects borrowed and adopted the traditions and ways of the fallen Persian empire
- Islamic architecture borrows heavily from Persian architecture.
- Characteristics: Tapered brick pillars Large arcades



Moorish - Alhambra

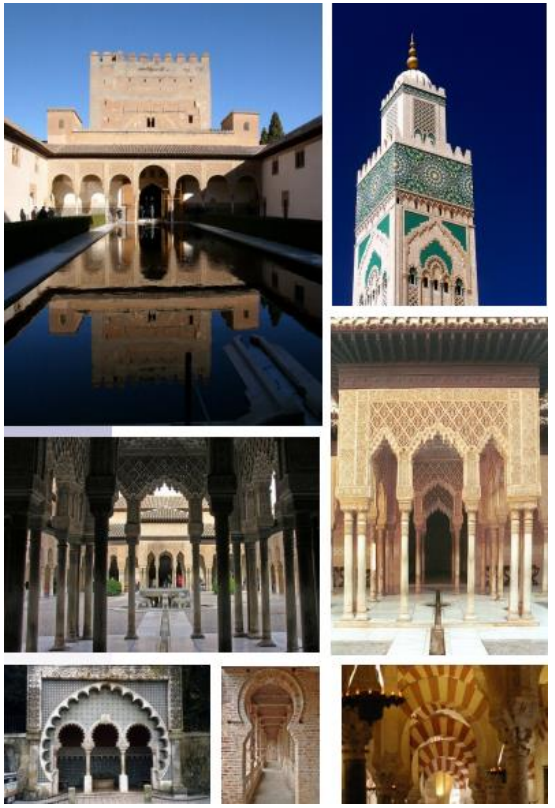
Great Mosque, Cordoba (785 AD)

- Peak of design: Alhambra, Granada (Palace/Fortress)
- Lasting impact on modern Spanish architecture.
- Characteristics:

  Foliage Motifs

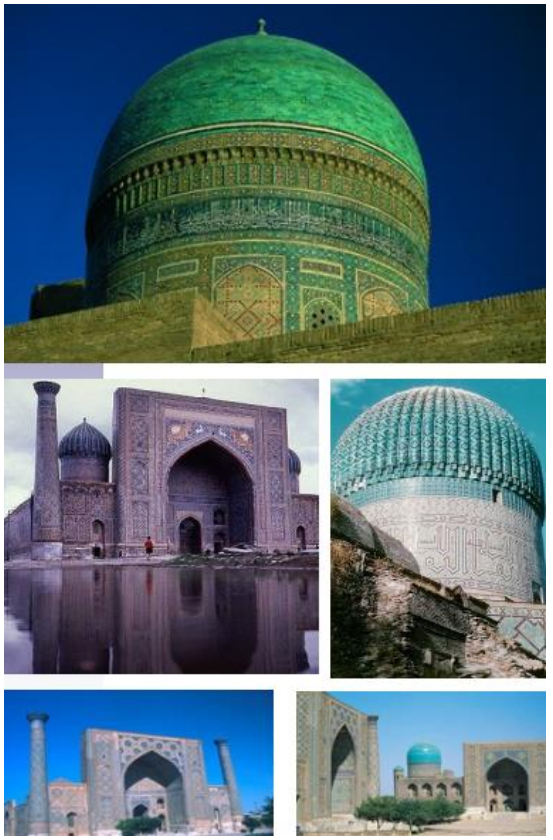
  Arches Glazed Tile work

  Central Fountain



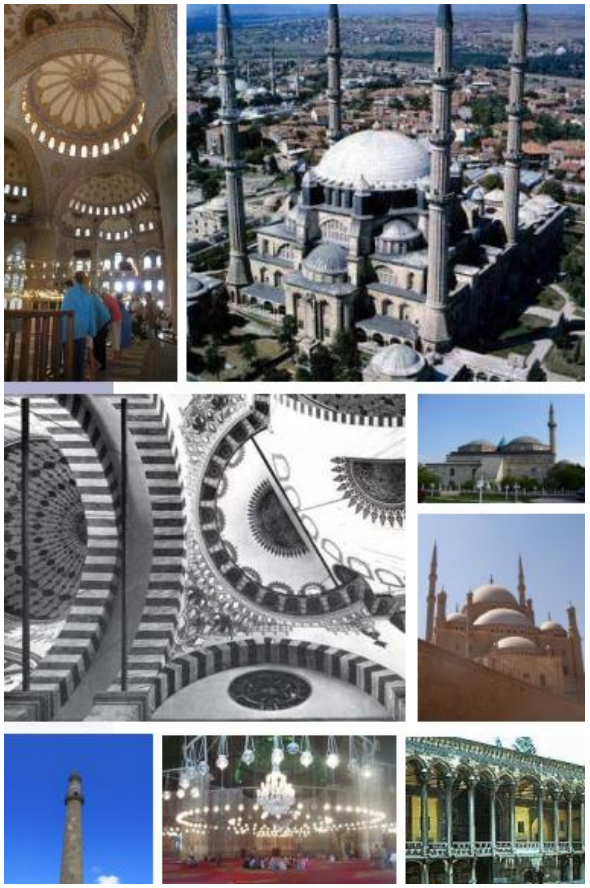
Turkistan - Samarkand

- Erected by Timur
- Samarkand, Herat, Meshed, Kazakhstan
- Derived from Persian influence
- Gave rise to Mughal design
- Characteristics:
  - Symmetry
  - Double Domes
  - Brilliant colors on the outside



Ottoman - Istanbul

- Byzantine, Persian and Syrian influence
- Vast inner spaces covered by seemingly weightless but massive domes
- Vaults, domes, semi-domes, columns
- Great architect – Sinan
- Note worthy architecture Suleiman Mosque



Fatimid & Mamluk- Cairo

- FATAMID High influence

- Turkistan

  Al-Azhar Mosque - 1st example

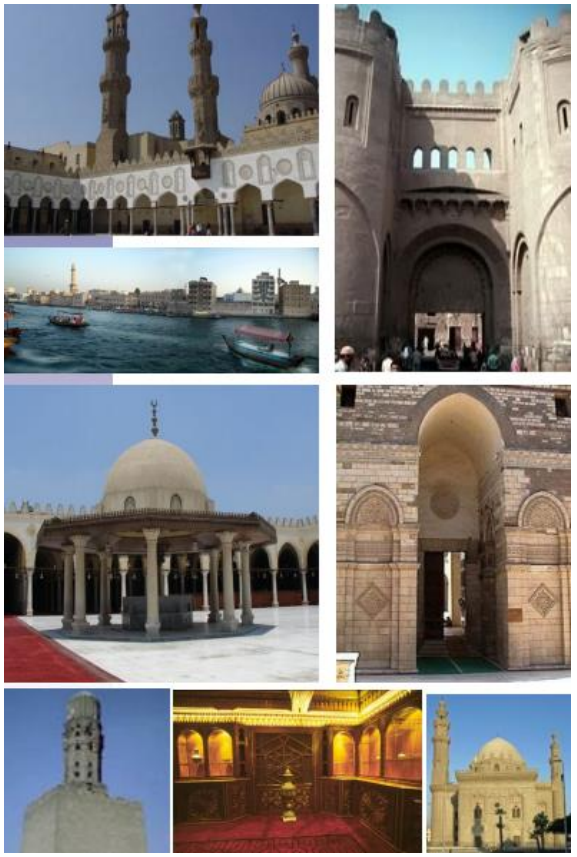
  Famous buildings

  - Al Azhar University
  - Al Hakim Mosque
  - Al Jame-al-Aqmer

- MAMLUK

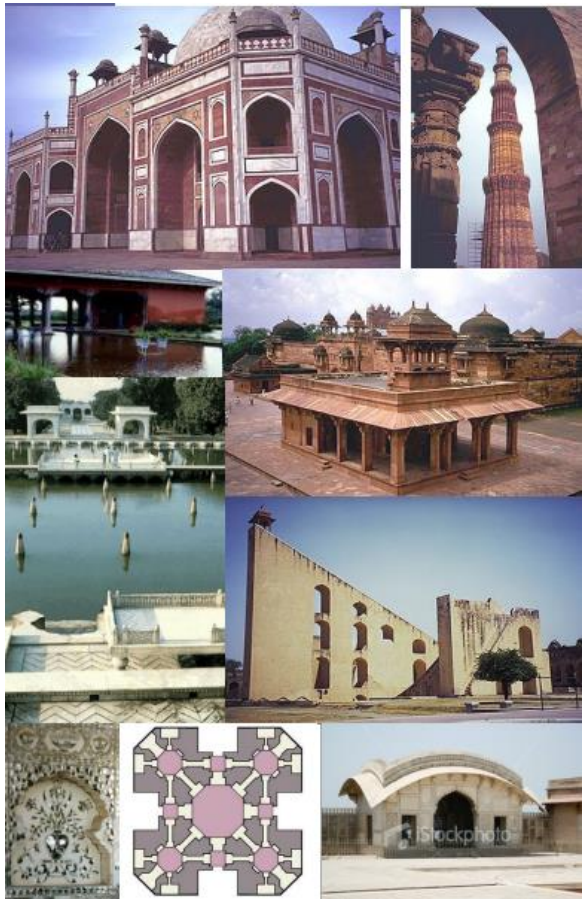
  Patterns of arts

  Art & Architecture at peak



Indo – Islamic (Mughal)

- Fusion: Persian, Arabic, Hindu
- Use marble & precious/semiprecious stone
- Showed power & influence over region
- Famous buildings:
  - Shalimar Gardens
  - Delhi Fort
  - Badshahi Masjid
  - Taj Mahal



Sino - Islamic

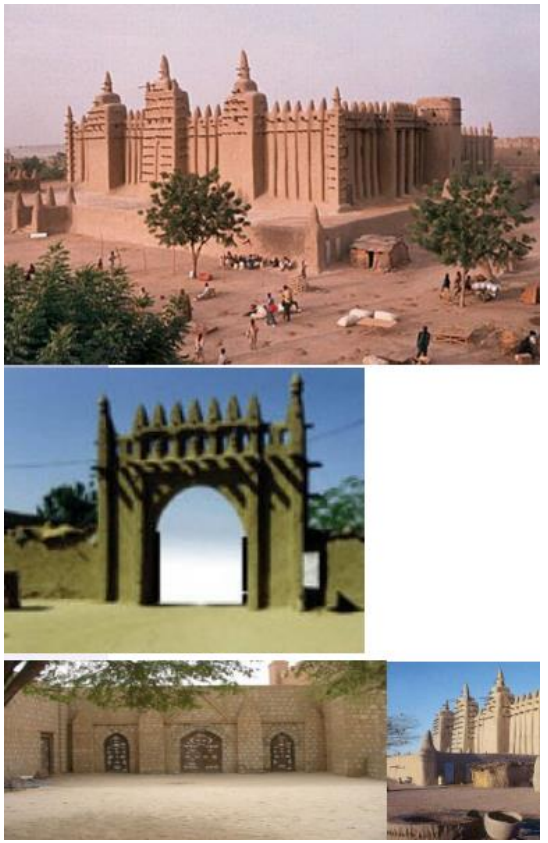
1 st Mosque in China: Grand Mosque of Xi'an

- Follow traditional Chinese form Buildings highly symmetrical but gardens highly asymmetrical
- Predominantly wooden construction – but some use of red or gray blocks



Afro Islamic

- Famous Structures in Africa:
  Grand Mosque of Djenne'
  Cairo Citadel Sankore
  Mosque in Timbutku
- Style highly influenced by Muslim traders to the region



Modern Islamic Architecture

- Burj Al-Arab Dubai, UAE
- Petronas Towers, Kuala Lumpur, Malaysia
- The Burj Dubai, UAE

Burj Al-Arab



Tallest Hotel – 1,053’ (1994)





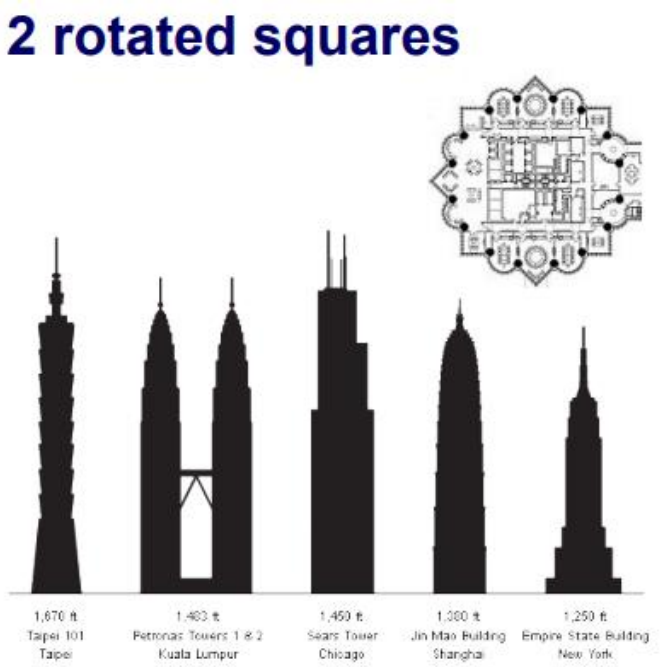






Petronas Towers







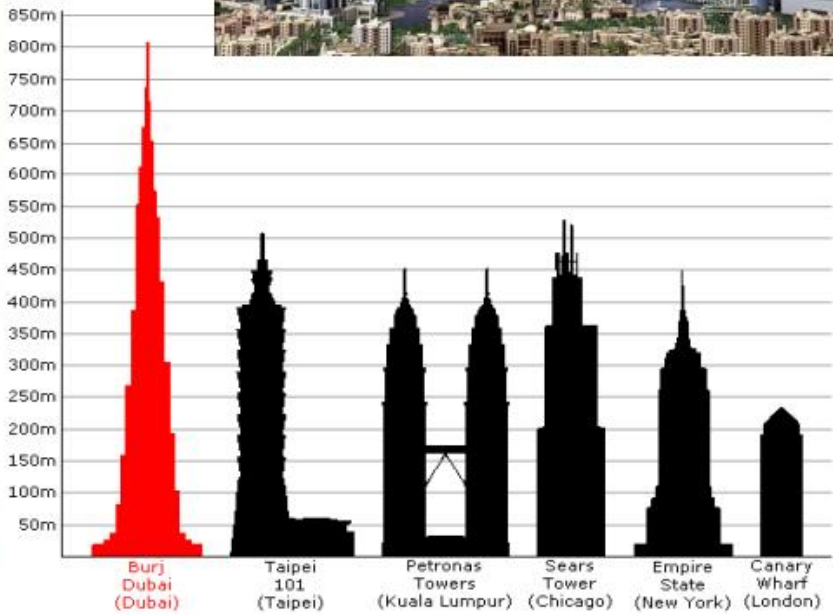


The Burj Dubai

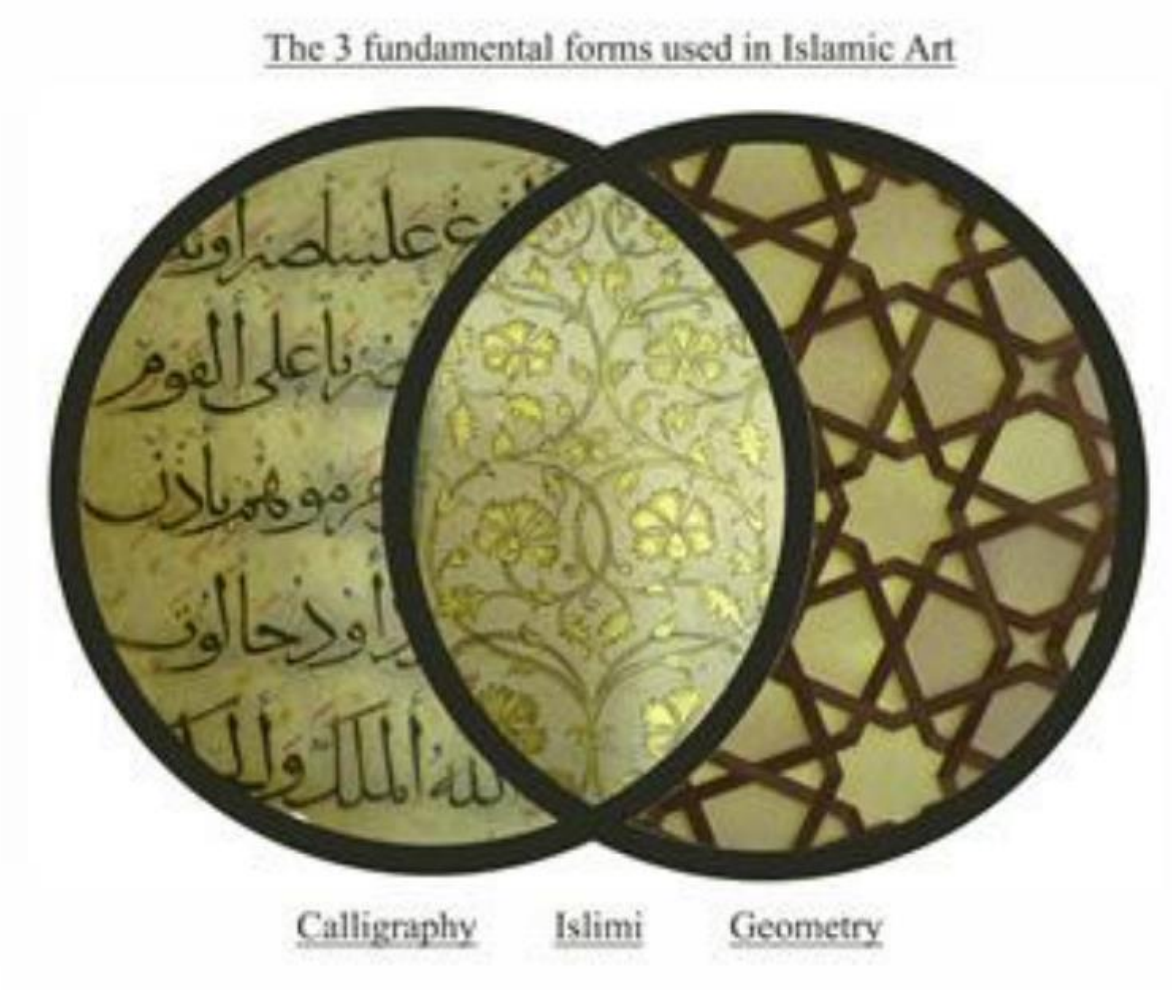


Abstracted desert flower pattern

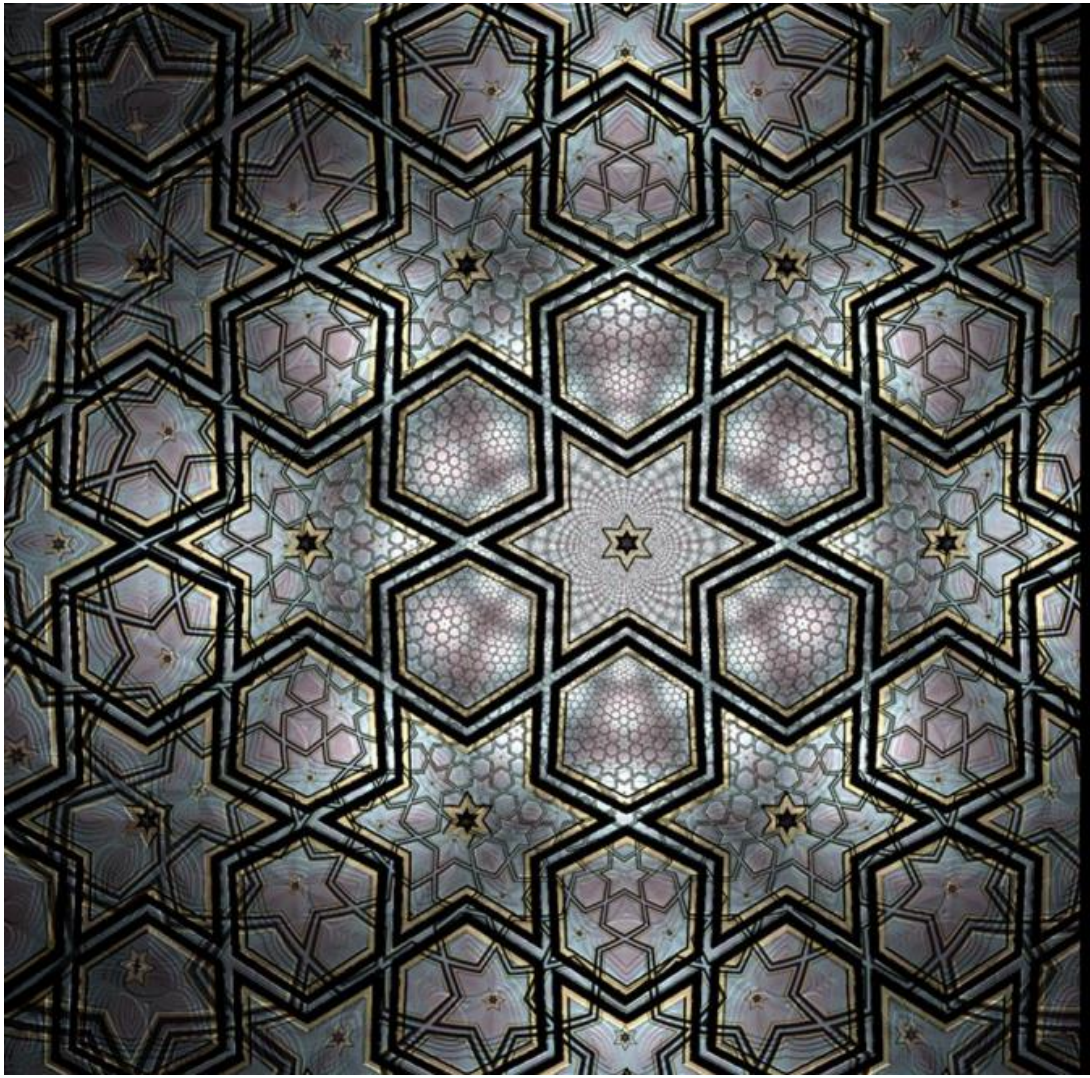




Islamic Decorative Elements



Islamic Decorative Elements



Geometric Pattern

These patterns exemplify the Islamic interest in **repetition, balance, symmetry and continuous generation of pattern.**

The integration of geometry with such **optical effects as the balancing of positive and negative areas**, a skillful use of **color and tone values.**

Islamic Decorative Elements



Arabesque (islimi)

Arabesque designs are biomorphic, **floral patterns** representing the underlying order and unity of **nature** with a great **deal of accuracy**. flowers and trees might be used as the **motifs for the decoration** of **textiles, objects and buildings**.

Islamic Decorative Elements



Calligraphy

Like other Islamic decoration, calligraphy is closely **linked to geometry**. The **proportions of the letters** are all governed by **mathematics**. **Inscriptions** are most often used as a frame along and around main elements of a building like portals and cornices.

Islamic Decorative Elements



Light

Light can add a **dynamic quality to architecture, extending patterns, forms** and designs into the dimensions of time.

And the combination of **light and shade** creates strong contrasts of planes and gives texture to sculpted stone, as well as stocked or brick surfaces.

Islamic Decorative Elements



Water

In hot Islamic climates, the water from courtyard pools and fountains **cools as it decorates**. Water can not only reflect architecture and multiply the decorative themes, it can also serve as a means of emphasizing the visual axes.